

AWS Glue Practical lab 3

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In this practical, I created an **ETL pipeline using Amazon S3 and AWS Glue**. The purpose of the task was to take employee and department data from S3, combine and clean the data, and then save the processed result into another S3 bucket.

I created three Amazon S3 buckets. The first two buckets were used as the **source bucket**, where I stored the original employee and department file. The third bucket was used as the **target bucket**, where the final processed data would be stored. Using separate source and target buckets helped me understand the difference between **raw data** and **processed data**.

- **Raw data** is data in its **original form**, exactly as it was collected. It hasn't been cleaned, changed, or organised yet.
- **Processed data** has been **cleaned, transformed, organised, or prepared** so it can be analysed

I uploaded emp_1.txt and department.txt into the two source S3 buckets. These files contained the information that I wanted AWS Glue to process. The employee file contained information such as employee ID, name, salary, address, location and email. The department file contained additional information such as department and designation.

The screenshot displays the AWS S3 console configuration for two buckets. The top section shows the configuration for the 'ana-s3-lab3-source' bucket, and the bottom section shows the configuration for the 'ana-s3-lab3-target' bucket. Both sections include a 'Bucket name' field, a 'Copy settings from existing bucket - optional' section, and a 'Choose bucket' button. The format for the bucket name is specified as 's3://bucket/prefix'.

Recent regional namespaces recommended:
General purpose buckets created in your account Regional namespace

Bucket name | [Info](#)

ana-s3-lab3-source

Bucket names must be 3 to 63 characters and unique within the global namespace

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

General purpose buckets created in your account Regional namespace

Bucket name | [Info](#)

ana-s3-lab3-target

Bucket names must be 3 to 63 characters and unique within the global namespace

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

I created a new job using **AWS Glue Visual ETL**. Visual ETL allowed me to create the data pipeline by adding and connecting different nodes.

AWS Glue was used to perform the ETL process:

Extract – read the original data from S3.

Transform – join, clean and change the data.

Load – save the processed data into the target S3 bucket.

Getting started

SageMaker Unified Studio 

ETL jobs

Visual ETL

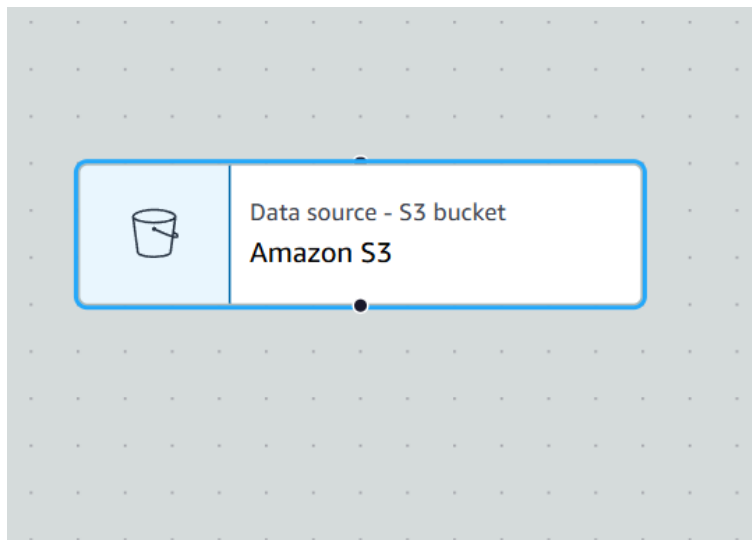
Notebooks

Job run monitoring

Data Catalog tables

Data connections

I added two **Amazon S3 data source nodes** because I was working with two datasets. I selected **CSV** as the data format and comma as the delimiter. This tells Glue that the values in the files are separated by commas. I also selected **First line of source file contains column headers** so that Glue could recognise values such as emp_id, name and salary as column names instead of normal records. I used **Infer Schema** so that AWS Glue could identify the fields in the datasets and their data types.



S3 URL

 s3://ana-s3-lab3-source/ 

[View](#) 



Recursive

Read files in all subdirectories.

Data format

CSV

Delimiter

Comma (,)

Escape character - *optional*

Enter a character to use for escaping

The character which immediately follows is used as-is, except for a small escapes (\n, \r, \t, and \0)

Quote character

Double quote (")



First line of source file contains column headers



Records in source files can span multiple lines



[Infer schema](#)

Unsaved job found

We found an unsaved job, do you

s3://ana-s3-lab3-sou X View ↗

☒ Recursive
Read files in all subdirectories.

Data format
CSV

Delimiter
Comma (,)

Escape character - optional
Enter a character to use for escaping

The character which immediately follows is used as-is, except well-known escapes (\n, \r, \t, and \0)

Quote character
Double quote (")

☒ First line of source file contains column headers

☐ Records in source files can span multiple lines

↻ Infer schema



Data source - S3 bucket

Amazon S3

✓

I added a **Join transformation** to combine information from the two S3 sources.

I selected an **Inner Join**, which returns records where the join condition finds a match between both datasets.

I configured the join condition using the matching fields from the two datasets. This allowed Glue to understand which records from the employee data should be combined with records from the department data.

The purpose of the Join was to create one dataset containing information that was previously stored separately.

Name

Join

Node parents

Choose which nodes will provide inputs for this one.

Choose one or more parent node

Amazon S3
S3 - DataSource

Amazon S3
S3 - DataSource

Join type

Select the type of join to perform.



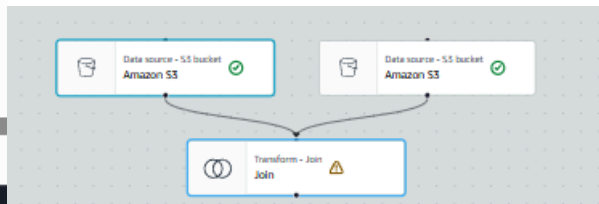
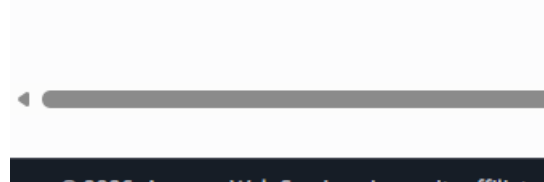
Inner join

Select all rows from both datasets that meet the

Join conditions

Select a field from each parent node for the join condition

Add condition



Choose which nodes will provide inputs for this one.

Choose one or more parent node

Amazon S3
S3 - DataSource

Amazon S3
S3 - DataSource

Join type

Select the type of join to perform.



Inner join

Select all rows from both datasets that meet the join condition.

Join conditions

Select a field from each parent node for the join condition.

Amazon S3

emp_id



=

Amazon S3

department



Add condition

After joining the datasets, I added a **Drop Fields transformation**.

This transformation was used to remove a field that I did not need in the final dataset. Removing unnecessary fields makes the output cleaner and means that only useful information continues through the pipeline.

Drop Fields removes a **column**, rather than deleting the entire record.

I also worked with the **Drop Null Fields transformation** and selected empty strings as values that could represent null data.

Null or empty values represent missing information. This transformation can be used to remove fields that contain no useful information because their values are null or empty.

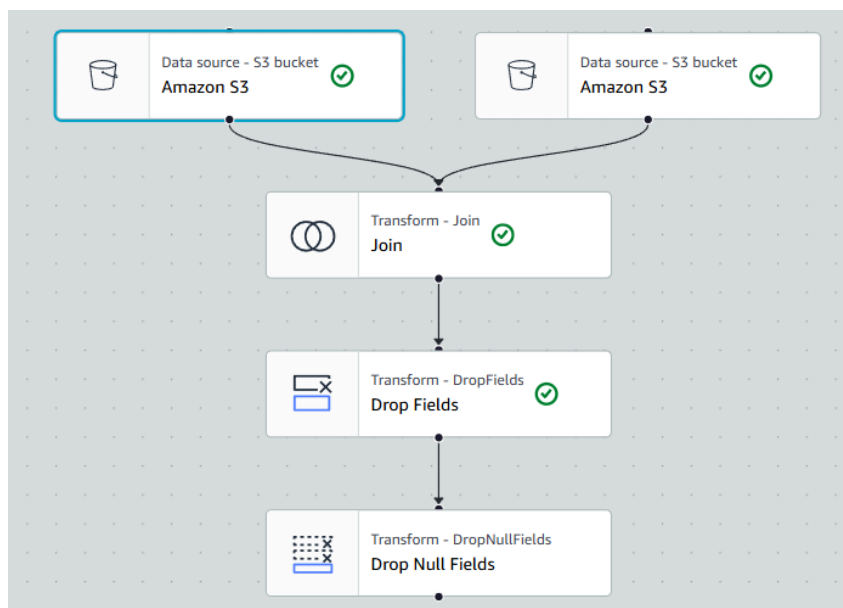
This helped me understand how AWS Glue can be used for basic **data cleaning**.

I used the **Change Schema (Apply Mapping)** transformation to control the structure and data types of the output.

Most of the fields remained as strings, but I changed the salary field from **string to double**.

I made this change because salary represents numeric data. Using a numeric data type makes the field more suitable for calculations such as total salary, average salary, minimum salary or maximum salary.

Change Schema can also be used to rename fields, change data types or remove fields when preparing data for the target.



Name

Drop Null Fields

Node parents

Choose which nodes will provide inputs for this node

Choose one or more parent node

Drop Fields



DropFields - Transform

DropNullFields | Info

Remove fields or columns where all the values are

Choose additional values that represent a null value

☒ Empty String (" " or "")

☐ "null" String

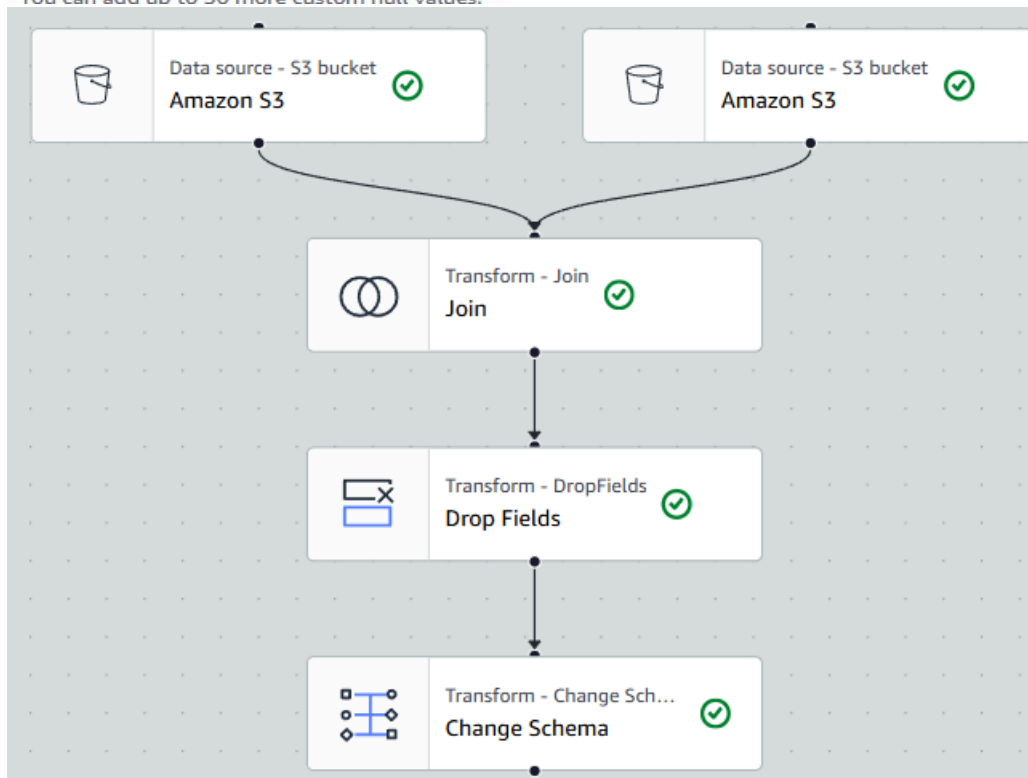
☐ -1 Integer

Add custom null value

Specify custom null values by entering the value

Add new value

You can add up to 50 more custom null values.



Change Schema (Apply mapping)

Source key	Target key	Data type
emp_id	emp_id	string ▼
name	name	string ▼
salary	salary	double ▼
address	address	string ▼
loc	loc	string ▼
email	email	string ▼
department	departme	string ▼
designation	designatic	string ▼

After completing the transformations, I configured an **Amazon S3 target**.

The target is where AWS Glue writes the final processed data after completing all of the transformations.

I selected an **IAM role** for the Glue job. The IAM role gives AWS Glue permission to access the AWS resources required by the job.

For example, Glue needs permission to read the files from the source S3 bucket and write the processed files into the target S3 bucket.

S3 Target Location

Choose an S3 location in the format `s3://bucket/prefix/object/` with a trailing slash (/).

Data Catalog update options [Info](#)

Visual

Script

Job details

Runs

Data quality

Schedules

Version Control

Basic properties

Info

Name

Ana Glue Lab3

Description - optional

Descriptions can be up to 2048 characters long.

IAM Role

Role assumed by the job with permission to access your data stores. Ensure that this role has permission to your Amazon S3 sources, targets, temporary directory, scripts, and any libraries used by the job.

AWSGlueServiceRole-Ana-salesproject

Create new role

Type

The type of ETL job. This is set automatically based on the types of data sources you have selected.

throughout the job run. Requires Glue 3.0 or later.

Requested number of workers

The number of workers you want AWS Glue to allocate to this job

2

Job runs (1/1)

Info

Filter job runs by property

	Run status	Retries
	Succeeded	0

ObjectsMetadataPropertiesPermissionsMetricsManagementFile s

Objects (4)



Copy S3 URI



Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects

 Find objects by prefix

☐	Name	Type	Last modified
☐	 run-1786458325543-part-block-0-r-00000-snappy.parquet	parquet	August 11, 202
☐	 run-1786458325543-part-block-0-r-00001-snappy.parquet	parquet	August 11, 202
☐	 run-1786458325543-part-block-0-r-00002-snappy.parquet	parquet	August 11, 202
☐	 run-1786458325543-part-block-0-r-00003-snappy.parquet	parquet	August 11, 202

Conclusions:

From this practical, I learned how different AWS services and Glue transformations work together in a basic data engineering pipeline.

I learned that **Amazon S3 stores the data**, while **AWS Glue processes and transforms the data**. I also learned how to combine two datasets using an Inner Join, remove unnecessary fields, handle empty data, change data types and save the final result into another S3 bucket.

Overall, this practical helped me understand the basic ETL process and how raw data can be transformed into cleaner and more useful data using AWS Glue.